CAMO

Designing explainable camouflage patterns from real environments

Can a pattern be designed from the places where it will be used?

Reference photographs → analysis → pattern generation → human and computer validation → optimization



1. From photographs to a first pattern

The baseline is procedural, transparent, and reproducible

INPUT

Representative landscape photos for forest, desert, mountain, snow, urban, jungle, taiga, and tundra.

ANALYSIS

Dominant colors, luminance, local contrast, texture, and relevant terrain pixels.

OUTPUT

A 500 × 500 finite-palette pattern with irregular connected regions.

Water and sky can be excluded or down-weighted when they are not relevant to the intended terrain.

2. Apply the pattern to a human figure

A flat square is not enough: scale, folds, seams, and garment pieces matter

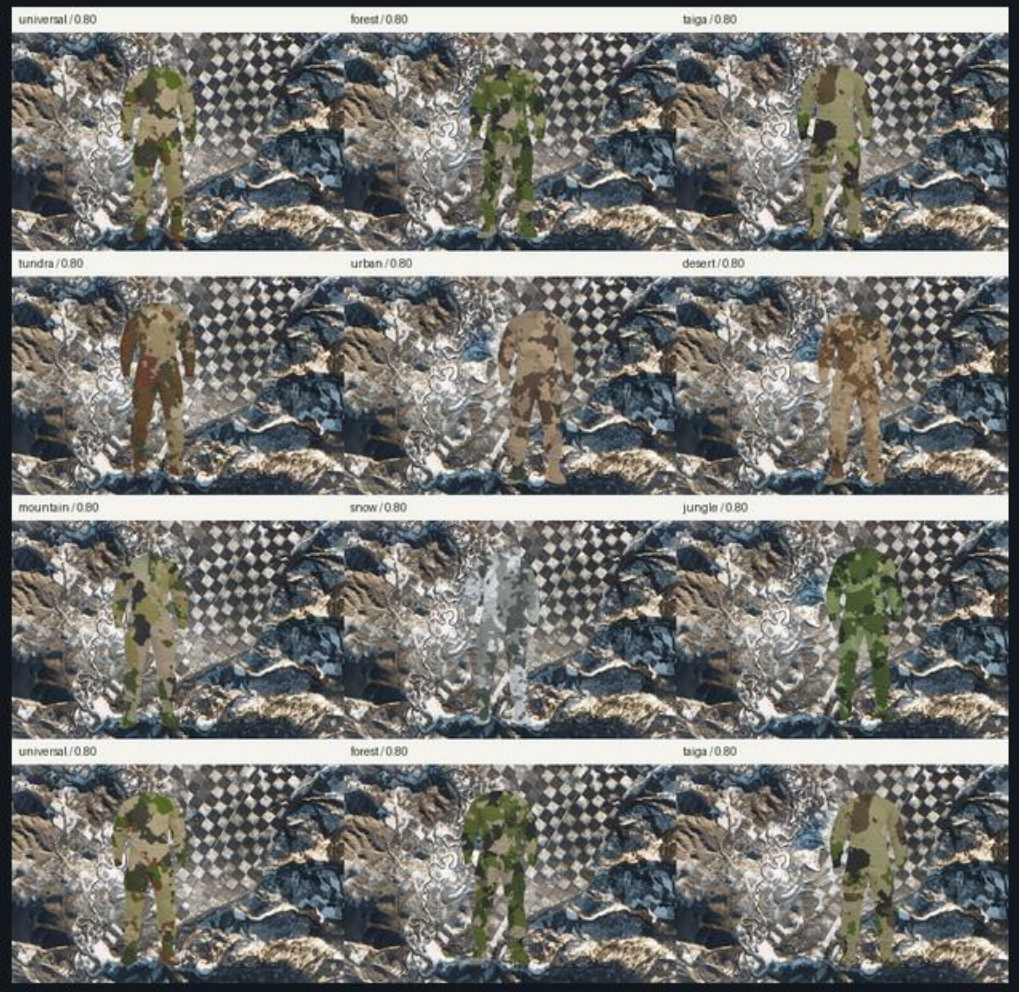


WHAT WE CHECK

- overly large patches
- excessive dark colors
- stripes or repetitive edges
- behavior at different scales

3. Test recognition in real environments

The figure is placed on terrain at different positions and scales



HUMAN TEST

Short exposures and randomized locations measure how quickly observers notice the figure.

COMPUTER METRICS

Color, luminance, edge, and visibility proxies compare candidate patterns.

RULE

Prefer terrain over obvious sky or water regions.

4. Improve the pattern with evidence

Optimization balances several objectives instead of fitting one photograph

A → Generate candidates

B → Score multiple scenes

C → Compare human results

D → Keep the best balanced pattern

photos → initial pattern → human/computer evaluation → adjusted pattern → repeat

Human CSV results are currently validation evidence. They are not yet used to train a neural network.

5. Where the neural network fits

SegFormer improves scene understanding; it does not generate the camouflage



LANDSCAPE PHOTO

The pretrained model assigns semantic labels such as vegetation, terrain, sky, and water.

RELEVANT PIXELS

The pipeline can focus color and texture extraction on the terrain that matters.

LIMIT

This is not custom neural-network training and not proof of real-world concealment.

Current status

A working research prototype with a reproducible workflow

- 9 environment profiles
- 80 documented evaluation images
- standard and weighted multi-objective generation
- optional neural segmentation
- uniform mockups and blinded human testing
- JSON configurations, tests, and reports



Next milestone: learn a ranking model from a larger human-test dataset while keeping the procedural baseline.